

scope

1. Inventory Management

1.1 Product Master

- Product creation
- Product categorization
- SKU management
- Unit of measurement
- Product status

1.2 Stock Management

- Current stock
- Available stock
- Reserved stock
- Damaged stock
- In-transit stock

1.3 Inventory Monitoring

- Stock-level monitoring
- Minimum stock monitoring
- Safety-stock monitoring
- Reorder-point monitoring

1.4 Inventory Classification

- Fast-moving products
- Slow-moving products
- Dead stock
- Overstocked products
- Critical products

1.5 Expiry Management

- Expiry-date tracking
 - Near-expiry detection
 - FEFO support
 - Expiry-risk alerts
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2. Demand & Forecast Management

2.1 Demand Data

- Historical sales
- Daily demand
- Weekly demand
- Monthly demand
- Seasonal demand

2.2 Forecasting

- Demand prediction
- Forecast horizon
- Forecast confidence
- Forecast updates

2.3 Demand Anomaly Detection

- Unexpected demand increase
- Unexpected demand decrease
- Demand spikes
- Demand drops

2.4 Forecast Monitoring

- Actual vs forecast
 - Forecast error
 - Demand deviation
 - Forecast accuracy
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3. Procurement & Purchase Order Management

3.1 Procurement

- Purchase requisition
- Supplier selection
- Purchase recommendation

- Purchase quantity calculation

3.2 Purchase Orders

- PO creation
- PO status
- PO tracking
- Expected delivery date
- Delayed PO detection

3.3 Lead-Time Management

- Supplier lead time
- Historical lead time
- Lead-time variability
- Lead-time changes

3.4 Procurement Risk

- Late delivery
 - Partial delivery
 - PO cancellation
 - Supplier dependency
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4. Supplier Management

4.1 Supplier Master

- Supplier profiles
- Products supplied
- Contact information
- Contract information

4.2 Supplier Performance

- On-time delivery
- Fill rate
- Quality performance
- Lead-time consistency

4.3 Supplier Risk

- Delay risk
- Reliability risk
- Dependency risk
- Capacity risk

4.4 Alternative Suppliers

- Supplier alternatives
 - Alternative-source recommendations
 - Supplier comparison
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5. Warehouse & Stock Distribution

5.1 Warehouse Management

- Warehouse master
- Warehouse capacity
- Warehouse inventory

5.2 Stock Allocation

- Warehouse allocation
- Regional allocation
- Store allocation

5.3 Inventory Transfer

- Transfer recommendation
- Transfer request
- Transfer approval
- Transfer tracking

5.4 Distribution Risk

- Warehouse shortage
- Warehouse overcapacity
- Location imbalance
- Transfer delays

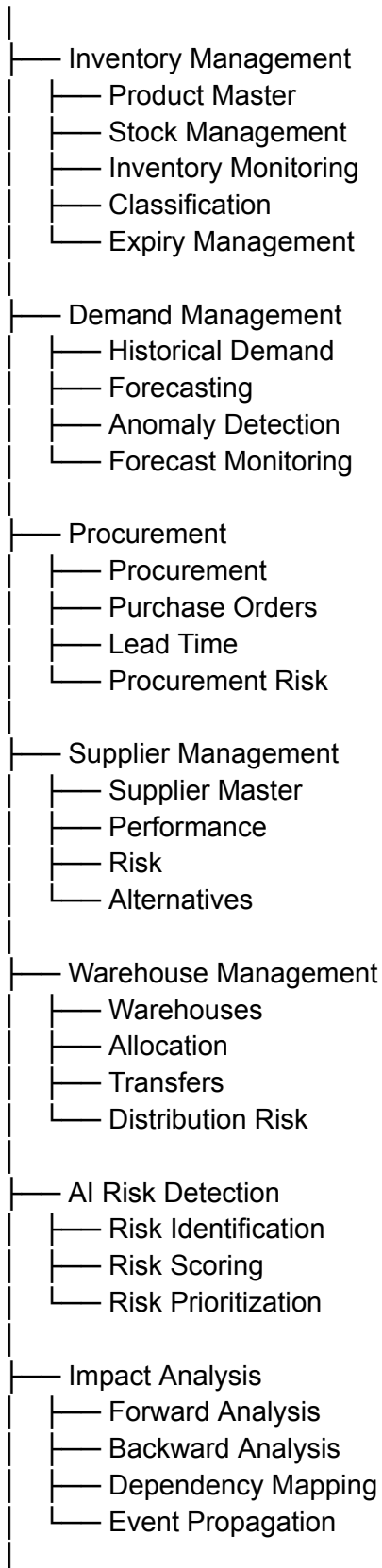
Primary User Personas

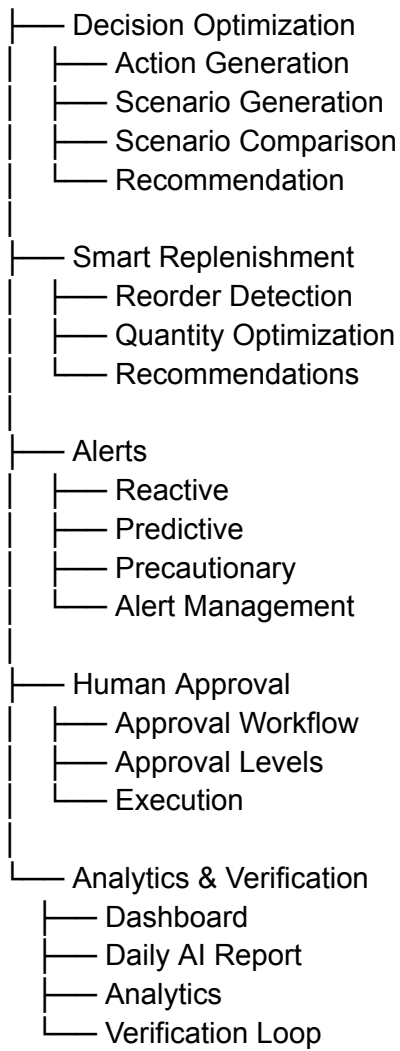
the main users are:

1. **Inventory Manager**
2. **Procurement Manager**
3. **Supply Chain Manager**
4. **Warehouse Manager**
5. **Operations Manager**
6. **Business/Admin**
7. **System/AI Administrator**

scope map

start





Story Map

Activity	User Goal	User Stories
Monitor	Understand current situation	View inventory, demand, suppliers, POs
Identify	Find risks	Detect stockout, overstock, supplier and expiry risks
Understand	Understand why	View risk causes and dependencies
Analyze	Understand consequences	View forward/backward impact
Decide	Choose action	Generate and compare scenarios
Approve	Control decisions	Approve/reject/modify AI recommendations
Execute	Perform action	Create PO, transfer stock, expedite shipment
Verify	Confirm outcome	Compare expected vs actual results
Improve	Learn from outcomes	Track historical performance and model feedback

backlog

Priority	Meaning	Approx. Tandex Scope
Must Have	Required for MVP	29
Should Have	Important next layer	17
Could Have	Enhancement	12
Won't Have	Explicitly excluded now	9

MUST HAVE

These are essential for the product to function as an intelligent inventory system.

ID	Requirement
M01	User authentication
M02	Role-based access control
M03	Product/SKU master
M04	Supplier master
M05	Warehouse master
M06	Inventory tracking
M07	Stock-level monitoring
M08	Purchase-order tracking
M09	Supplier-delay detection
M10	Demand data ingestion
M11	Stockout-risk detection

M12	Overstock detection
M13	Risk scoring
M14	Risk prioritization
M15	Forward impact analysis
M16	Basic backward impact analysis
M17	Event propagation
M18	Replenishment recommendation
M19	Decision scenario generation
M20	Scenario comparison
M21	AI recommendation
M22	Human approval
M23	Approved-action execution
M24	Action verification
M25	Risk recalculation
M26	Alert management
M27	Dashboard
M28	Daily AI stock report
M29	Audit trail

These form the **minimum intelligent Tandex product**.

5.2 SHOULD HAVE

Important features that significantly improve the product but are not required for the first functioning version.

ID	Requirement
S0 1	Demand forecasting
S0 2	Forecast accuracy tracking
S0 3	Expiry-risk detection
S0 4	Slow-moving inventory detection
S0 5	Dead-stock detection
S0 6	Supplier performance scoring
S0 7	Alternative supplier recommendations
S0 8	Warehouse transfer recommendations
S0 9	Safety-stock optimization
S1 0	Lead-time prediction
S11	Precautionary alerts
S1 2	Advanced backward root-cause analysis
S1 3	Advanced scenario simulation
S1 4	Cost optimization
S1 5	Alert escalation
S1 6	Advanced analytics

5.3 COULD HAVE

Useful enhancements that can be implemented after the core system becomes stable.

ID	Requirement
C01	Natural-language inventory queries
C02	Conversational AI assistant
C03	Voice-based interaction
C04	Advanced demand anomaly detection
C05	Automated supplier negotiation suggestions
C06	What-if simulation
C07	Advanced predictive analytics
C08	Mobile application
C09	Custom dashboard builder
C10	Advanced visualization
C11	External market-data integration
C12	Automated report customization

5.4 WON'T HAVE — Current Release

These are deliberately excluded from the initial release.

ID	Requirement
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- W01 Fully autonomous procurement
- W02 Fully autonomous supplier switching
- W03 Fully autonomous high-value purchasing
- W04 Fully autonomous inventory transfers
- W05 Autonomous contract negotiation
- W06 Autonomous financial transactions
- W07 Full ERP replacement
- W08 Global multi-enterprise deployment
- W09 Fully autonomous supply-chain control

sprint log

1. Sprint 1 Goal

Build the foundational platform with authentication, role-based access, master data, and basic inventory management.

Sprint 1 flow

